Personalized Fashion Recommendation System

Urvi Sharma
urvi.sharma.v@gmail.com

Prof. G P Sajeev
sajeevgp@am.amrita.edu

November 11, 2021

Amrita School Of Engineering
Amrita Vishwa Vidyapeetham

Outline

Recap

Introduction

Related Work

Research Gap

Proposed Method

Algorithms

Validation

Action Plan

Recap

Sugestions:

- Dataflow Explaination
- Improved PageRank
- Tags use-case

Background

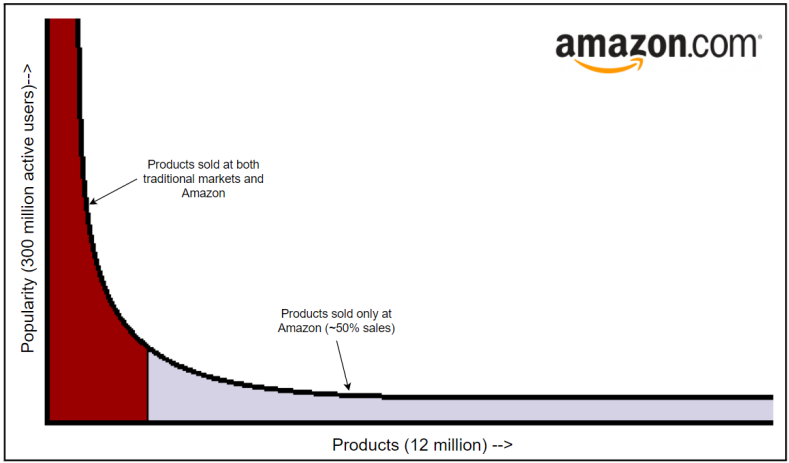


Figure 1: Amazon's Long Tail Phenomenon

Clothing Recommender System

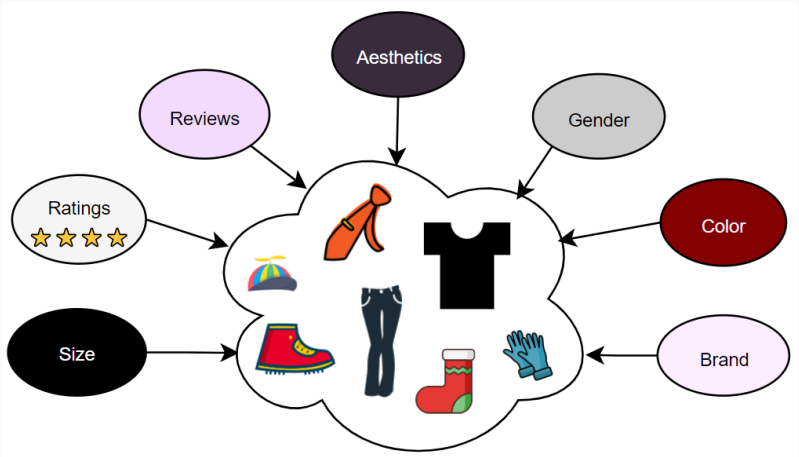


Figure 2: Factors that affect buyer's decision making while buying clothing articles and hence, justify the need for feature based recommender system.

Objectives

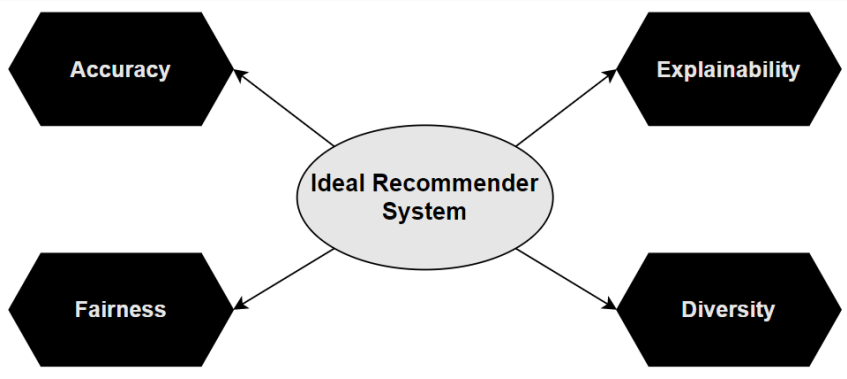


Figure 3: Objectives of an ideal Recommender System [Gao et al., 2021]

Stages

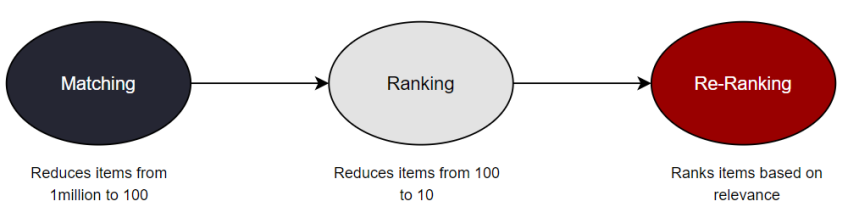


Figure 4: General steps in Product Recommender systems [Gao et al., 2021]

Techniques

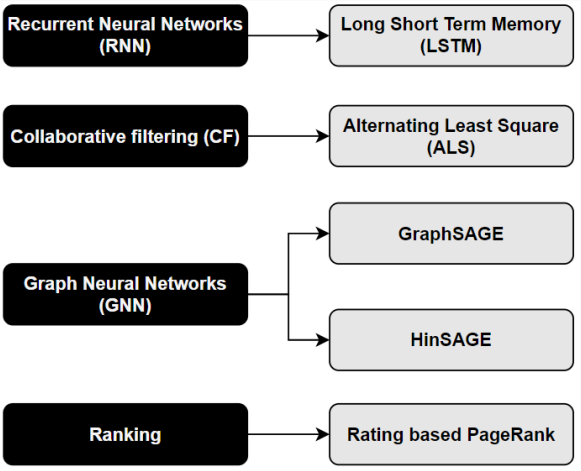


Figure 5: Algorithms used in this project

Related Work

Paper	Method	Merit	Demerit
[He and McAuley, 201	One-Class Collaborative Filtering	Visual Preference Analysis	Cold Start
[Cheng et al., 2018]	Aspect-Aware Latent Factor Model	Feedback Association	Visual Context
[Yan et al., 2019]	Data Augmentation	Cold Start	Trend Awareness
[Wen et al., 2018]	Apriory Algorithm	Monotonicity	High Dimensional Data
[Ying et al., 2018]	GCNN	Deep Learning	Computationally Expensive

Table 1: Summary of reviewed papers.

Related Work-Summary

	He et al.,	Cheng et al.,	Ostuni et al.,	Yan et al.,	Wen et al.,	Ying et al.,
Image Processing	Yes	No	No	Yes	No	No
Text Processing	Yes	Yes	No	No	No	No
Graph Algorithm	No	No	Yes	Yes	Yes	Yes
Collaborative Filtering	Yes	No	No	No	No	No
Time Sensitive	Yes	No	No	No	No	No
Sequence Aware	No	No	No	No	No	No

Table 2: Summary of reviewed papers based on Techniques used.

Research Gap and Challenges

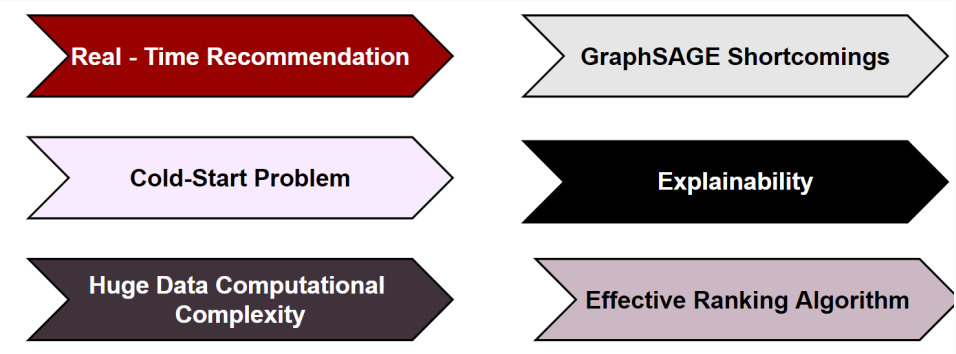


Figure 6: Major challenges in research

Workflow

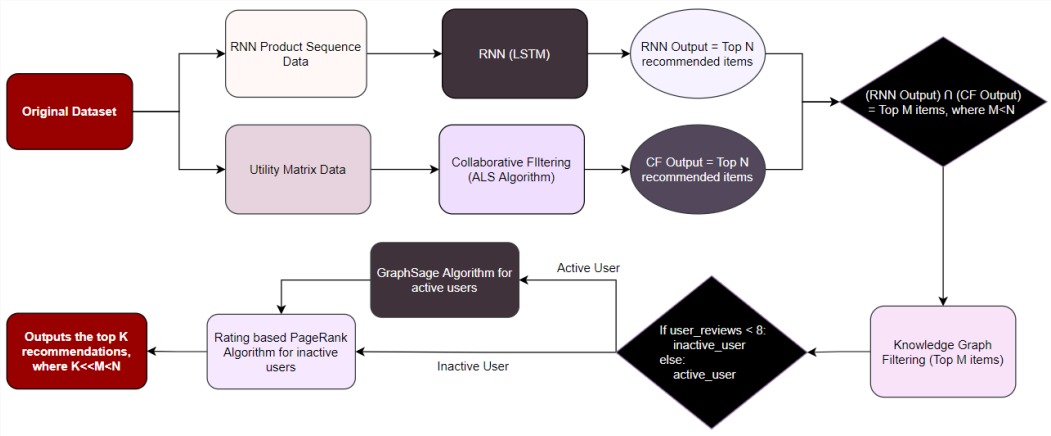


Figure 7: Model Structure

Recurrent Neural Network

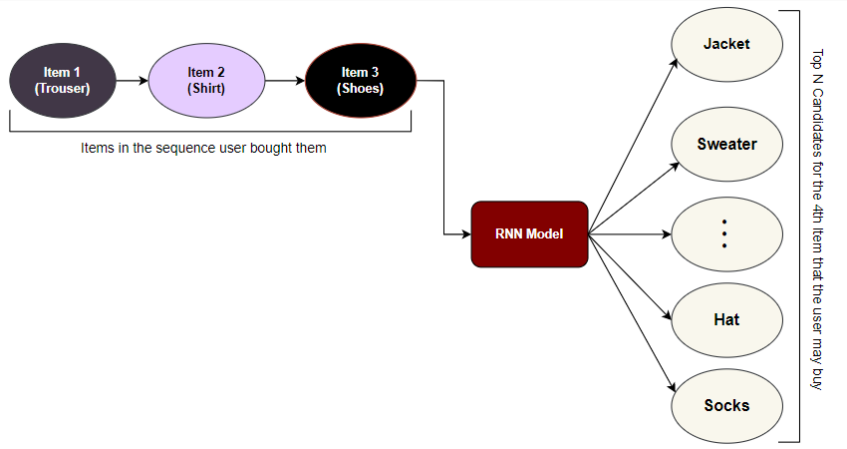


Figure 8: Given the product sequence for user history the RNN model will predict the possible N items user could buy next.

Graph Recommender System

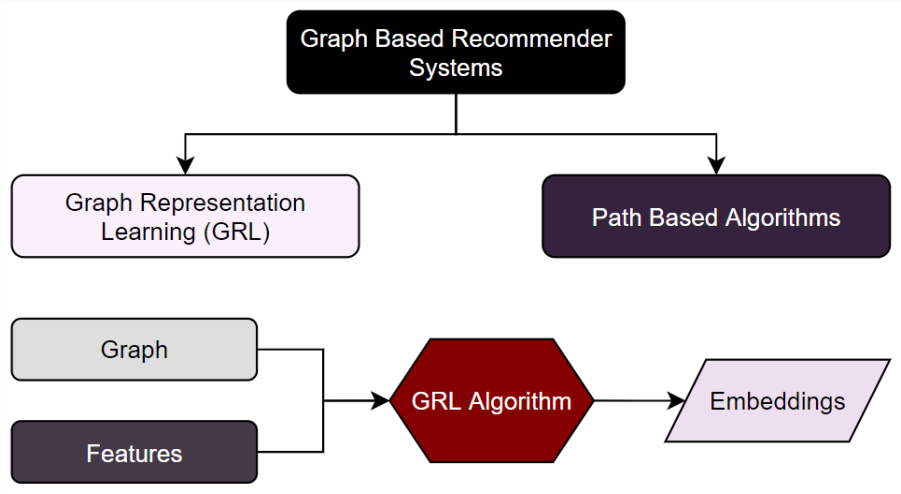


Figure 9: Why use Graph Representation Learning?

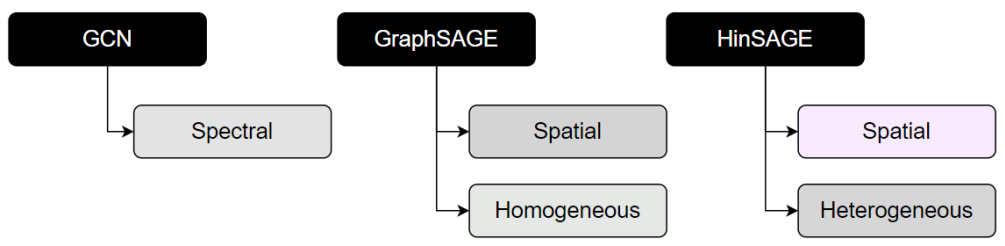


Figure 10: GNN Model Comparison

Resulting Graph

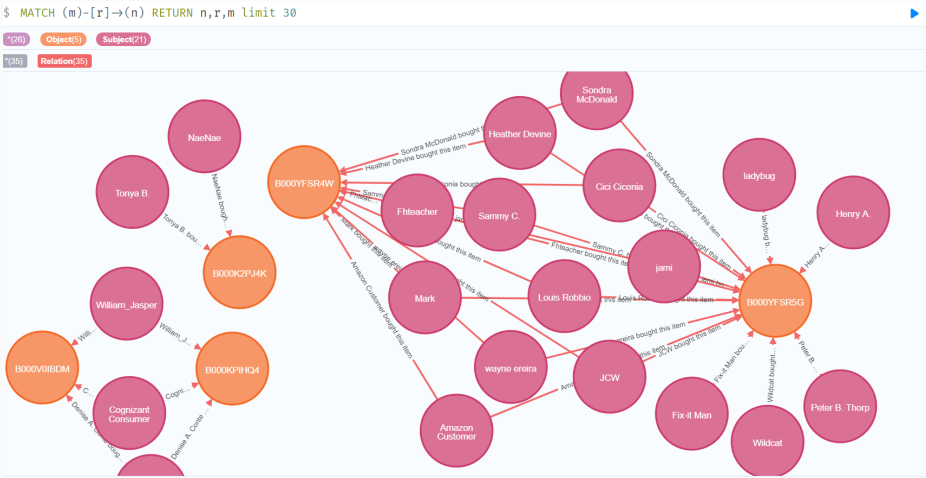


Figure 11: Resulting Knowledge Graph

Evaluation

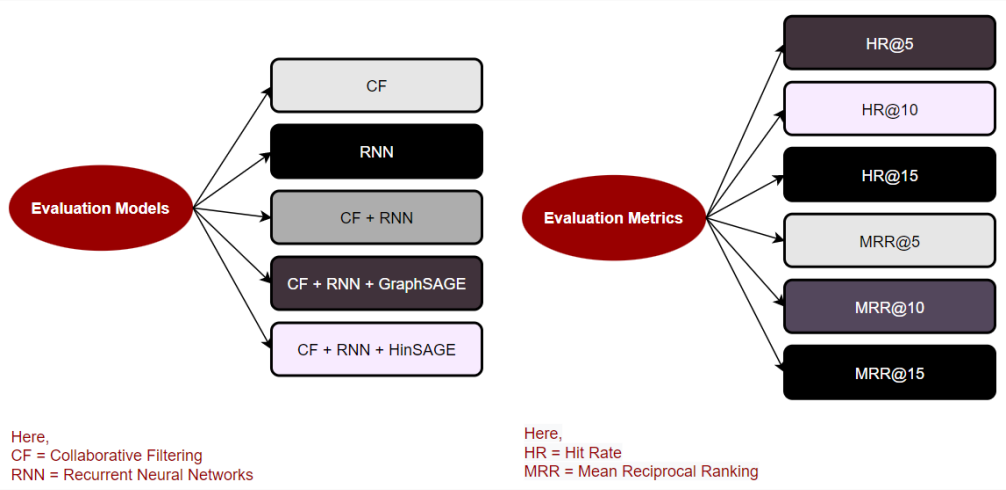


Figure 12: Comparative analysis with baseline models

Metrics Used

Hit Rate (HR): Whether the item user buys next is present in the recommended list.

Mean Reciprocal Ranking (MRR): Where the item user buys next is present (ranked) in the recommended list. It measures the ranking of the recommender system. Best case scenario is when relevant items are present in the top of the list.

Rank	User 1	User 2
1	Hat	Top
2	Shirt	**Sweater**
3	**Shoes**	Jeans
4	Socks	Jumper
5	Gloves	Cap

Table 3: MRR Evaluation Data

$$\text{MRR} = \frac{1}{|X|} \sum_{i=0}^{|X|} \frac{1}{\text{Rank}_i}$$

Hence,

$\text{MRR} = 1/2\ (1/3 + 1/2) = 0.416$

Action Plan

- GraphSAGE Baseline Implementation: Completed
- HinSAGE Baseline Implementation: Completed
- ALS Implementation: 30th-Nov
- RNN Implementation: 30th-Nov
- Research Paper Documentation: December Onwards

References i

Cheng, Z., Ding, Y., Zhu, L., and Kankanhalli, M. (2018).
Aspect-aware latent factor model: Rating prediction with ratings and reviews.
In *Proceedings of the 2018 world wide web conference*, pages 639–648.

Gao, C., Zheng, Y., Li, N., Li, Y., Qin, Y., Piao, J., Quan, Y., Chang, J., Jin, D., He, X., et al. (2021).
Graph neural networks for recommender systems: Challenges, methods, and directions.
arXiv preprint arXiv:2109.12843.

References ii

He, R. and McAuley, J. (2016).
Ups and downs: Modeling the visual evolution of fashion trends with one-class collaborative filtering.
In *proceedings of the 25th international conference on world wide web*, pages 507–517.

Wen, Y., Liu, X., and Xu, B. (2018).
Personalized clothing recommendation based on knowledge graph.
In *2018 International Conference on Audio, Language and Image Processing (ICALIP)*, pages 1–5. IEEE.

References iii

Yan, C., Chen, Y., and Zhou, L. (2019).
Differentiated fashion recommendation using knowledge graph and data augmentation.
Ieee Access, 7:102239–102248.

Ying, R., He, R., Chen, K., Eksombatchai, P., Hamilton, W. L., and Leskovec, J. (2018).
Graph convolutional neural networks for web-scale recommender systems.
In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 974–983.